Exponentials & Logarithms

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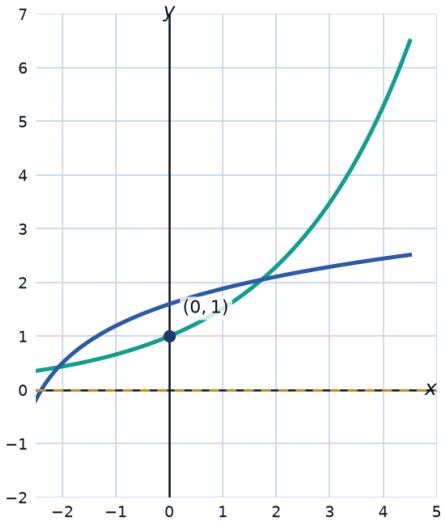
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Natural Exponential & Logarithm

An interval of width 200 containing a root must be reduced to width at most 1 using repeated bisection. Find the minimum number of bisections required.

2 [Foundational] • Target: Exponential Equations with Different Bases

Evaluate $\int_{1}^{2} (2x + 5)^{1}\, dx$.



3 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Evaluate $\int_{0}^{\ln(8)/4} 5e^{4x}\, dx$.

4 [Foundational] • Target: Exponential Growth and Decay Modeling

Evaluate $\int_{0}^{\ln(4)/2} e^{2x}\, dx$.

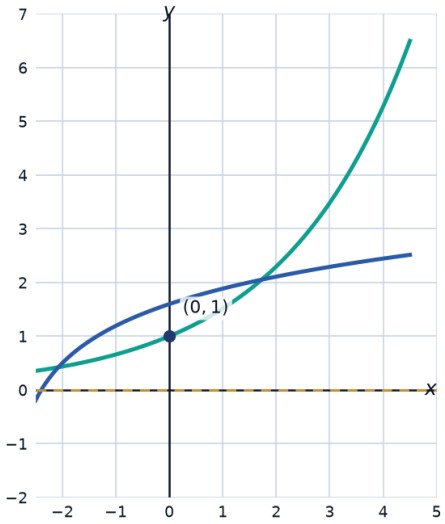
5 [Foundational] • Target: The Natural Exponential & Logarithm

Solve $|4x + 12| > 6$. What is the smaller boundary value separating the two regions?

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6 [Foundational] • Target: Exponential Equations with Different Bases

Find $\int \frac{2x}{(x+2)(x-1)}\,dx$.



7 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Evaluate $\int_0^5 (x-12)^5\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential Growth and Decay Modeling

Evaluate $\int_0^4 (4x-5)^4\,dx$.

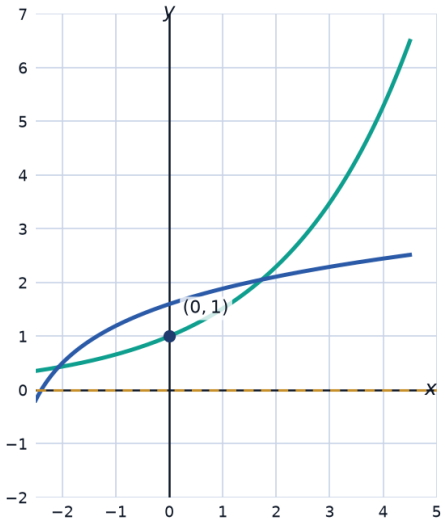
9 [Intermediate] • Target: The Natural Exponential & Logarithm

Evaluate $\int_{-1}^0 (3x-8)^1\,dx$.

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10 **[Intermediate]** • **Target: Exponential Equations with Different Bases**

Evaluate $\int_{-2}^{1}(5x+8)^4\,dx$.



11 **[Intermediate]** • **Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)**

Solve $e^{2x} - 23e^x + 130 = 0$ for x. Give the larger solution in exact form.

12 **[Intermediate]** • **Target: Exponential Growth and Decay Modeling**

Solve $e^{2x} - 21e^x + 110 = 0$ for x. Give the larger solution in exact form.

13 **[Intermediate]** • **Target: The Natural Exponential & Logarithm**

Solve $e^{2x} - 28e^x + 187 = 0$ for x. Give the larger solution in exact form.

14 **[Intermediate]** • **Target: Exponential Equations with Different Bases**

Solve $e^{2x} - 22e^x + 120 = 0$ for x. Give the larger solution in exact form.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

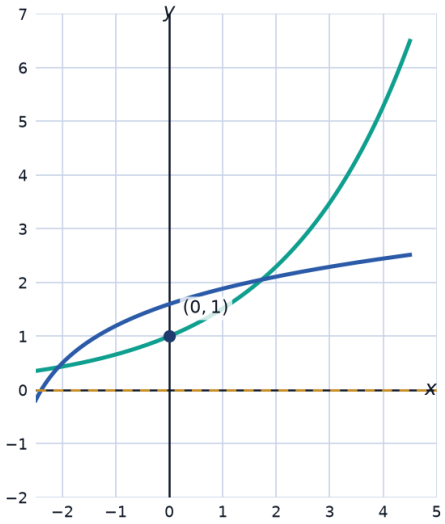
15 **[Challenging]** • **Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)** • **Cross-Lesson: 3.01 Advanced Trigonometry**

Solve $e^{2x} - 39e^x + 360 = 0$ for x. Give the larger solution in exact form.

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16 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 4.01 Differentiation

Solve $e^{2x} - 22e^{x} + 112 = 0$ for x. Give the larger solution in exact form.



17 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 5.01 Integration

Solve $e^{2x} - 13e^{x} + 30 = 0$ for x. Give the larger solution in exact form.

18 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 6.01 Sequences & Series

Solve $e^{2x} - 24e^{x} + 108 = 0$ for x. Give the larger solution in exact form.

19 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 17e^{x} + 42 = 0$ for x. Give the larger solution in exact form.

20 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Solve $e^{2x} - 19e^{x} + 48 = 0$ for x. Give the larger solution in exact form.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Natural Exponential & Logarithm

Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \frac{1 + \ln 5}{2}$

b. $x = \frac{\ln 5 - 1}{2}$

c. $x = \ln\left(\frac{5}{2}\right) + 1$

d. $x = \frac{\ln 6}{2}$

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22 [Foundational] • Target: The Natural Exponential & Logarithm

Solve $\ln(3x+2) - \ln(x) = \ln 4$ for $x > 0$.

a. $x = 2$

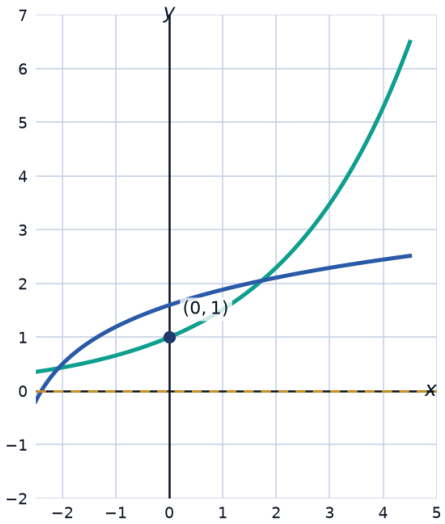
b. $x = 1$

c. $x = 4$

d. $x = 1/2$

23 [Foundational] • Target: The Natural Exponential & Logarithm

Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.



a. $x^3 + 2x$

b. $3x + 2x = 5x$

c. $x^3 + e^{2x}$

d. $3x^2 + 2$

24 [Foundational] • Target: Exponential Equations with Different Bases

Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 2}{\ln 3 - \ln 2}$

b. $x = \frac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \frac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \frac{1}{\ln(3/2)}$

25 [Foundational] • Target: Exponential Equations with Different Bases

Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

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26 [Foundational] • Target: Exponential Equations with Different Bases

Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10

b. 0.852

c. 1.21

d. 0.945

27 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. Gradient $= \ln b$, Intercept $= \ln a$

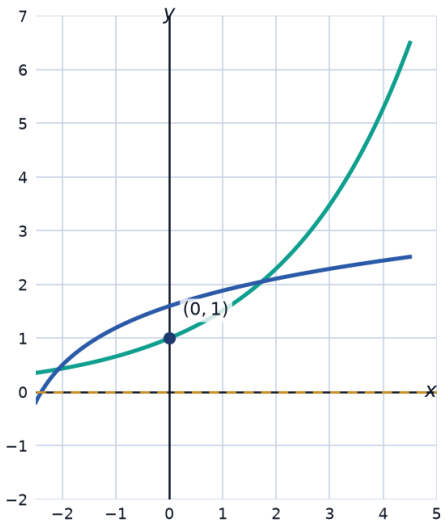
b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

28 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.



a. $k = 4, n = 2.5$

b. $k = 2.5, n = 4$

c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)**

Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.

a. $b = e^{0.8} \approx 2.226$

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

30 **[Intermediate]** • **Target: Exponential Growth and Decay Modeling**

The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years

b. 20.0 years

c. 10.0 years

d. 15.5 years

31 **[Intermediate]** • **Target: Exponential Growth and Decay Modeling**

A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \frac{\ln 4}{3}$

b. $k = \frac{\ln 2}{3}$

c. $k = \frac{4}{3}$

d. $k = \ln\left(\frac{4}{3}\right)$

32 **[Intermediate]** • **Target: Exponential Growth and Decay Modeling**

A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min

b. 15.2 min

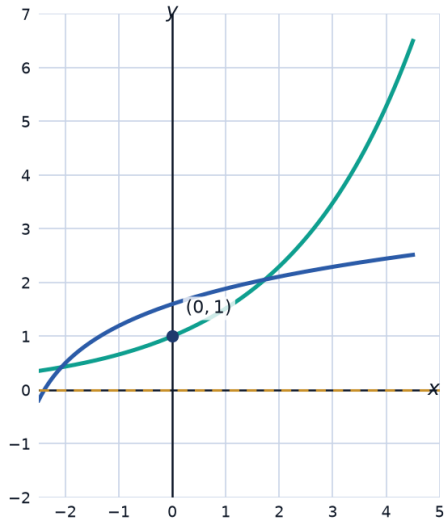
c. 10.5 min

d. 18.0 min

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33 [Intermediate] • Target: The Natural Exponential & Logarithm

Solve $e^{2x} - 14e^{x} + 33 = 0$ for x. Give the larger solution in exact form.



a. $\ln 11$

b. $\ln 14$

c. $\ln 33$

d. $\ln 3$

34 [Intermediate] • Target: Exponential Equations with Different Bases

Solve $e^{2x} - 25e^{x} + 144 = 0$ for x. Give the larger solution in exact form.

a. $\ln 16$

b. $\ln 9$

c. $\ln 25$

d. $\ln 144$

35 [Intermediate] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Solve $e^{2x} - 26e^{x} + 120 = 0$ for x. Give the larger solution in exact form.

a. $\ln 6$

b. $\ln 120$

c. $\ln 20$

d. $\ln 26$

36 [Intermediate] • Target: Exponential Growth and Decay Modeling

Solve $e^{2x} - 15e^{x} + 44 = 0$ for x. Give the larger solution in exact form.

a. $\ln 15$

b. $\ln 11$

c. $\ln 4$

d. $\ln 44$

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37 [Intermediate] • Target: The Natural Exponential & Logarithm

Solve $e^{2x} - 27e^{x} + 140 = 0$ for x. Give the larger solution in exact form.

a. $\ln 20$

b. $\ln 27$

c. $\ln 140$

d. $\ln 7$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 6.01 Sequences & Series

Solve $e^{2x} - 28e^{x} + 52 = 0$ for x. Give the larger solution in exact form.

a. $\ln 52$

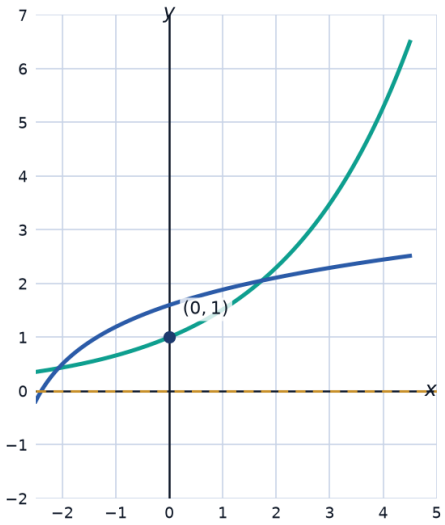
b. $\ln 26$

c. $\ln 2$

d. $\ln 28$

39 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 23e^{x} + 42 = 0$ for x. Give the larger solution in exact form.



a. $\ln 21$

b. $\ln 42$

c. $\ln 2$

d. $\ln 23$

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40 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Solve $e^{2x} - 20e^{x} + 91 = 0$ for x. Give the larger solution in exact form.

a. $\ln 91$

b. $\ln 20$

c. $\ln 13$

d. $\ln 7$

41 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 9.01 Complex Numbers

Solve $e^{2x} - 7e^{x} + 10 = 0$ for x. Give the larger solution in exact form.

a. $\ln 2$

b. $\ln 10$

c. $\ln 7$

d. $\ln 5$

42 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Solve $e^{2x} - 7e^{x} + 12 = 0$ for x. Give the larger solution in exact form.

a. $\ln 12$

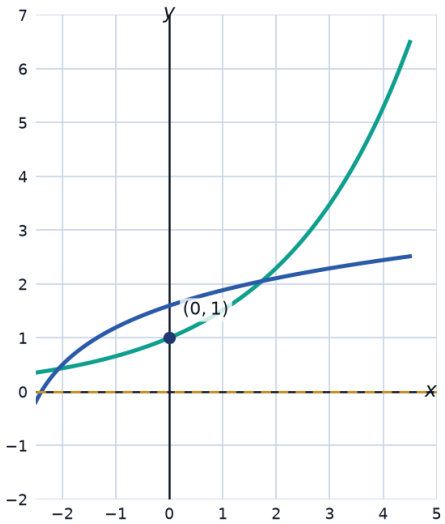
b. $\ln 3$

c. $\ln 4$

d. $\ln 7$

43 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 10.01 Numerical Methods

Solve $e^{2x} - 13e^{x} + 42 = 0$ for x. Give the larger solution in exact form.



a. $\ln 6$

b. $\ln 42$

c. $\ln 7$

d. $\ln 13$

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44 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 11.01 Differential Equations

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

a. $x = 11$ only

b. $x = 11$ and $x = -5$

c. $x = 5$ only

d. $x = 16$ only

45 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 12.01 Poisson Distribution

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

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